Ratul Hasan

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EDUCATION

Bangladesh Army University of Engineering & Technology (BAUET) *Natore, Bangladesh*
B.Sc in Computer Science and Engineering | CGPA: 3.82/4.0 *2022 2026*

Thesis Title: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimers Disease

Govt. Majid Memorial City College *Khulna, Bangladesh*
Higher Secondary School Certificate | GPA: 5.0/5.0 *2019 2021*

EXPERIENCE

ELITE Research Lab LLC | *Remote* *Queens, NY, USA*

- ***Research Assistant*** *Oct 2026 Present*
  Supervisor: Md Kishor Morol, PhD Student, Cornell University
  PI: S M Musfequr Rahman, PhD Student, Tampere University, Finland
  - Working on a team journal project mapping attacker-controllable IoT metadata as an indirect prompt-injection surface for LLM home agents
  - Measuring which Home Assistant device and integration text fields reach its built-in LLM conversation agents, and how often injected instructions trigger unauthorised physical actions or leak private state in a simulated home
  - Evaluating how provenance tagging, actuating-tool gating and action confirmation reduce attack success, and their cost in benign-task utility
- ***Research Intern*** *Jun 2026 Sep 2026*
  Supervisor: Md Kishor Morol, PhD Student, Cornell University
  PI: Shakib Sadat Shanto, PhD Student, University of Maine, USA
  - First-authored a systematic review of 82 studies on LLMs in mental health, appraising methodological rigour and risk reporting (hallucination, bias, safety, human evaluation) with a 24-field extraction instrument
  - Studied catastrophic forgetting in federated continual learning for mental health text classification, separating label-space shift from domain shift on Reddit stress and depression data
  - Benchmarked seven training regimes (FedAvg, FedProx, EWC, replay, centralized and joint baselines) across 10 clients and three seeds with a MiniLM encoder

BAUET Computer Society, Dept. of CSE | ***Programming Instructor*** *Natore, Bangladesh*
Jul 2024 Dec 2025

- Conducted programming classes for undergraduate students covering C, C++ and basic data structures.
- Led coding exercises and programming contest preparation sessions to strengthen students' problem-solving and analytical skills.
- Explained programming concepts to students of varying skill levels and supported the development of their programming fundamentals.

Lab AR | ***Software Engineering Intern*** *Dhaka, Bangladesh*
Jul 2025 Aug 2025

- Led a team project to develop a web-based Property Management System.
- Contributed to backend development using Laravel (PHP) and collaborated on feature implementation.

PUBLICATIONS

Conference Publications

Quantum-Resistant FOTA: End-to-End Decentralized Firmware Updates for IoT Using Blockchain and CRYSTALS-Dilithium *2025*
Ratul Hasan, Md. Momenul Haque, Redoanul Haque, Yousuf Oley and Ruhani Akter. *Proceedings of the 12th International Conference on Next Generation Computing, Communication, Systems and Security (NSysS '25)*. ACM, New York, NY, USA, 110–114. doi.org/10.1145/3777555.3777569

A Gas-Optimized Blockchain Framework for Scalable Document Verification Using Dynamic Access Control and IPFS *2026*
Ratul Hasan, Samiha Farjana, Yousuf Oley, Md. Ohiduzaman Pranto and Md Arik Rayhan. *International Conference on Power, Electronics, Communications, Computing and Intelligent Infrastructure (PECCII 2026)*. IEEE, 2026. doi.org/10.1109/PECCII70991.2026.11662071

SplitSafe Ultra: A Moving-Target Defense Framework for Entropy-Aware Encrypted File Persistence on IPFS *Expected 2026*

Md. Ahnaf Muhaimin, **Ratul Hasan**, Yousuf Oley, Samiha Farjana, Bristi Rani Roy, Subrata Kumer Paul and Md. Ekramul Hamid. Accepted to the *5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*. To be published in IEEE Proceedings.

ActiVisionNet-Fusion: A Dual-Stream Explainable Deep Learning Framework for Driver Drowsiness Detection *Expected 2026*

Nasim Ahmed, **Ratul Hasan**, Subrata Kumer Paul, Dewan Nafiul Islam Noor, Md. Momenul Haque and Md. Ekramul Hamid. Accepted to the *5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*. To be published in IEEE Proceedings.

Journal Manuscripts (Under Review / In Submission)

BI-RNN: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimers Disease *Under Review*

Ratul Hasan, Md. Momenul Haque, Ananya Sarker. Under review in *Scientific Reports*, Submission Date: May 2026.

Entropy-Assured Post-Quantum Key Generation on IoT Edge Devices: Source Independence, Measured Min-Entropy and Signature Scheme Selection *Under Review*

Ratul Hasan, Md Arik Rayhan and Md Ohiduzaman Pranto. Under review in *PLOS ONE*.

Methodological Rigour and Risk Evaluation of Large Language Models in Mental Health: A Systematic Review *Under Submission*

Ratul Hasan, Mizanul Hoque, Moinul Hossain, Md Kishor Morol and Tze Hui Liew. Systematic Review, under submission to *Frontiers in Digital Health* (Digital Mental Health section).

Preprints

Disentangling Label-Space Shift from Domain Shift in Federated Continual Learning for Mental Health Text Classification *2026*

Ratul Hasan and Mizanul Hoque. *SSRN Preprint*, 2026. doi.org/10.2139/ssrn.7662638

COMPETITIVE PROGRAMMING & OTHERS

Online Judges

- Codeforces: Expert, max 1650 (**Ratul_Hasan**) | CodeChef: 3 Star, max 1676 (**ratulhasan2108**)

Programming Contests

- ICPC Asia Dhaka Regional Contestant (Team: BAUET_Rong_Answer)
- National Collegiate Programming Contest (JU): Rank 149 (Team: BAUET_PARADIGM)
- Inter-University Programming Contest (2x): Best Rank 45 at IUT (Team: BAUET_Twisted_Minds)
- Ranked 10th in Independence Day Programming Contest by Military Institute of Science & Technology
- Intra-University Programming Contest: Champion and 2nd Runner-up (Team), Runner-up (Solo)

Others

- 1st Runner-up, Intra-University Three Minute Thesis (3MT) Presentation

PROJECTS

LedgerSeal: A Blockchain Based Document Verification Framework using Dynamic Access Control and IPFS *Github*

- Developed a decentralized application (DApp) for constant-time document verification with dynamic access control on IPFS using a public blockchain. *Tools: React JS, Anvil EVM, Foundry, MetaMask, Pinata IPFS*

BAUET Project & Thesis Archive *Github*

- Developed a repository website using Django (both MVT & REST) where theses and projects of BAUET are represented. *Tools: HTML, Tailwind CSS, JavaScript, Django, PostgreSQL*

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL, Solidity
ML & Research: PyTorch, Hugging Face (Transformers)
Web & Blockchain: Django, Django REST Framework, Laravel, React, Foundry, IPFS
Tools & Databases: Git, GitHub, Docker, Linux, PostgreSQL, MySQL
Research Areas: Deep Learning, NLP & LLMs, LLM Security, Federated & Continual Learning, Computer Vision, Post-Quantum Cryptography, Blockchain & IoT Security

VOLUNTEERING & LEADERSHIP EXPERIENCE

- President, BAUET Computer Society (Advanced through five executive leadership positions)
- Problem Setter, Intra-University Programming Contest, BAUET
- Officer of Documentation, Hult Prize at BAUET